

FS 572: Fermentation Science & Technology

Lecture 7
Microbial Fermentation Systems
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Course Topics

- Industrial fermentation overview
- Yeast and bacterial systems
- Fermentation kinetics
- Process control
- Product quality assurance

Saccharomyces cerevisiae

- Most common industrial fermentation yeast
- High ethanol tolerance
- Efficient sugar metabolism
- Used in brewing, distilling, and bioethanol production

Fermentation Process

1. Raw material preparation
2. Mashing and saccharification
3. Fermentation
4. Distillation or downstream processing
5. Quality testing

Factors Affecting Fermentation

- Temperature
- pH
- Nutrient availability
- Oxygen exposure
- Microbial contamination

Laboratory Exercise

Compare fermentation performance of two yeast strains.
Record gravity, pH, and temperature daily.
Prepare a short technical summary of findings.